

Anthony Ge

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Education

Northwestern University | Evanston, IL

Expected June 2028

Bachelor of Science, Computer Science and Mathematics (Double Major)

GPA: 3.95/4.00

- Courses: Probability & Stochastic Proc, ML, Applied Statistical Learning, Data Structures & Algorithms, Real Analysis, Optimization

Experience

Amazon

Jun. 2026 – Sep. 2026

SDE Intern, Amazon Prime Video

Seattle, WA

- Building an event-driven DataQualityObserver Lambda for Prime Video catalog export incremental feeds
- Designing reusable observer engine, feed readers, and quality-check interfaces across EMBER/GLOBAL live and VOD feeds
- Implementing schema conformance & required-field checks over XML/JSONL feeds to detect malformed records and missing data
- Emitting CloudWatch metrics, alarms, and dashboard signals to surface feed-quality defects before partner escalation

MedSphere

Jan. 2026 – Mar. 2026

Software Engineer Intern, AI

Skokie, IL

- Built document-to-graph pipeline extracting pharma/biotech text/metadata into graph nodes and edges
- Implemented entity extraction and normalization (canonicalization, deduping, synonym resolution) for consistent schema at scale
- Built relation extraction and graph construction for confidence-weighted typed edges and source-backed explanations
- Productionized graph builds via async jobs for batch/incremental updates, plus logging/monitoring and stable APIs for consumers

Northwestern FinTech

Sep. 2024

Quantitative Developer

Evanston, IL

- Built a hybrid trading pipeline with a C++ core and Python orchestration
- Integrated REST/WebSocket market-data clients, normalized schemas, and persisted datasets for reproducible runs
- Developed a feature builder and real-time inference path (microstructure signals); exposed C++ modules to Python via pybind11
- Created an event-driven backtesting harness with fee/slippage modeling and CLI reporting; added logging, tests, and handling
- Optimized critical paths through profiling and low-latency I/O/memory tuning; packaged with CMake/Conan for 1-command setup

Awards

USACO Platinum (United States of America Computing Olympiad)

- Qualified to Platinum (highest division) after advancing through Bronze → Silver → Gold via rated contests

HackPrinceton 2025 Finalist

- Built an ML pipeline to trade Kalshi high-temperature weather markets across 8 major U.S. cities using intra-day directional signals
- Trained and validated a probabilistic bucketed-forecast model (LightGBM) on 1-minute observations plus NOAA forecasts with feature engineering and calibration to predict daily highs
- Implemented real-time inference on streaming MADIS 1-minute data with support for partial-day inputs, producing contract-level probabilities for execution/backtesting and performance monitoring

Projects/Competitions

Optiver's Trading-At-The-Close Competition

- Developed an ensemble of ML models (CatBoost, GRU, Transformer) with 25 engineered features to predict NASDAQ stock closes
- Performed extensive feature engineering like time-based aggregations, rank features, rolling means, and ratio features
- Adopted a day-based validation strategy (train on first 400 days, validate on last 81 days) that aligned closely with the leaderboard
- Integrated post-processing steps (weighted-mean subtraction) to refine predictions and further reduce error metrics

S&P 500 Arima Forecasting Model – Quantitative Research Project

- Harvested and processed 13,000+ daily S&P 500 closing prices from 1975–2025 to develop a robust time series forecasting model
- Used Augmented Dickey-Fuller test and ACF/PACF plots to assess stationarity and select parameters
- Leveraged automated grid search techniques (via auto_arima) to identify the best ARIMA (p, d, q) configuration
- Mean Average Percentage Error: 2.48%

VIX Probabilistic Forecasting via AR-GARCH (t-errors)

- Built a modular ETL → Modeling → Evaluation pipeline: loads/cleans 2010–present VIX, applies transforms, fits rolling AR(1)–GARCH(1,1), and outputs forecasts plus 80/90/95% intervals
- Single-command runner with config files, logging, error handling, and reproducible, versioned artifacts (CSV/plots)
- Implemented comparison harness and reporting (tables/plots) with numerical-stability safeguards during optimization
- Validation: beat Random Walk and EWMA ($\alpha=0.06$); test RMSE 2.10 / MAE 1.06; coverage $\sim 0.868/0.925/0.955$; avg widths 3.98/5.28/6.51

Skills

Languages: Python, C/C++, Java, MATLAB, TypeScript/JavaScript, SQL, Lua

Web & Back-End: Node.js, React.js, Express.js, Svelte, REST APIs, HTML/CSS

Tools/Libraries: PyTorch, TensorFlow, Git, GitHub Actions, AWS, Firebase, CatBoost, pandas, NumPy, pmdarima